Landmine Victims in Colombia: A Reproducible Data Workflow

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Overview

I built a reproducible Python workflow, in one Jupyter notebook, that loads, cleans, explores and charts Colombia's official register of people killed or injured by landmines and unexploded ordnance. The dataset is *Situación Víctimas Minas Antipersonal en Colombia* [Situation of Antipersonnel Mine Victims in Colombia], published by the Presidencia de la República de Colombia on the national open data portal: https://www.datos.gov.co/Inclusi-n-Social-y-Reconciliaci-n/Situaci-n-V-ctimas-Minas-Antipersonal-en-Colombia/yhxn-eqqw. I chose it because I have worked in humanitarian mine action and arms control for 25 years, as an explosive ordnance disposal technician and a chemical weapons inspector, and I want to find where artificial intelligence can help save lives in the field. The notebook, data, `requirements.txt` and README are in a public GitHub repository, https://github.com/travisrcarlson/ai-programming-foundations-project, and the cleaning functions are written to be reused in later capstone projects.

Dataset Description

The register holds one row for each recorded victim of antipersonnel mines, improvised explosive devices that act like them, and unexploded ordnance in Colombia since 1990 (Presidencia de la República de Colombia, 2024). The snapshot I used was downloaded on 18 October 2026; the publisher last updated the data in January 2024, and the snapshot covers 1990 to 2023. It has 12,390 rows and 18 columns, recording the department and municipality, rural or urban area, year and month, age group, ethnic group, condition (civilian or member of the security forces), outcome (killed or injured), sex, type of place and the victim's activity at the time. I focused on condition, outcome, age group, year, department and civilian activity. The file

has no blank cells, but missing values are written as "Sin información" in several spellings, and three column labels do not match their values: `estado` holds sex, `genero` holds the outcome and `tipoevento` holds the type of place. The data are published under the CC BY-SA 4.0 licence.

Workflow Description

The whole workflow runs top to bottom in `data_workflow.ipynb`, which keeps the code, its outputs and my explanations in one document. Danchev (2022) uses Jupyter notebooks this way to support a transparent, reproducible workflow, while noting that a notebook does not by itself guarantee reproducibility; the pinned environment and the data checksum described below cover that gap.

1. **Ingestion.** The notebook checks the raw CSV's SHA-256 checksum against the value recorded at download, then reads the file unchanged with pandas and shows its first rows and column types.
2. **Cleaning.** Six small functions, each with a docstring, run in order as a pipeline: correct the column labels, drop free-text location detail, turn "no information" into real missing values, store the official DANE codes as text, translate the categories through a data dictionary, and add date fields. Each step returns a new table, and the pipeline logs rows, columns and missing cells after every step. Checks then confirm that every victim was kept and every category is known, and the cleaned table (12,390 rows × 17 columns) is saved as a separate CSV.
3. **Exploratory analysis.** Four functions profile each column, count victims and their share by any grouping, give the share killed per group, and count victims per year by condition; a correlation check compares the two yearly series.

4. **Visualizations.** Four charts (Figures 1–4), each with a title, labelled axes and a short interpretation, are saved to the `figures/` folder.
5. **Summary.** A final section states what the data show, my assumptions, the limitations and open questions.

Key Decisions and Assumptions

Cleaning Choices

I kept the raw file exactly as downloaded and made every change in code, so anyone can rerun the path from raw data to results. This follows the advice to save data as originally generated and to record every processing step in scripts (Wilson et al., 2017). The other choices were:

- **Labels corrected from their values.** `genero` contains only "Herido" and "Muerto" (injured and killed), which are outcomes, not genders. The function checks the values before renaming and stops with an error if a future release changes them.
- **"Sin información" turned into missing values,** so unknowns are counted as unknown rather than as a category.
- **Codes, not names, as the key.** Twenty-one municipality names belong to more than one department, which is why the data hold 519 municipality codes but only 498 names.
- **Location detail dropped.** Free-text site names, some of them street addresses, and municipal-seat coordinates are not needed for analysis by municipality.
- **Repeated rows kept.** 3,858 rows (31%) repeat another row exactly. The file has no victim or event identifier, so several people hurt in the same accident produce identical rows, and deleting them would delete real victims (see Responsible Practice).

Assumptions

Each row is one victim. The ethnic-group value "no" means none or not recorded. Three spellings of illicit-crop eradication, and two of security-force duty, each describe one activity.

EDA Focus

I looked at who was harmed (civilians or security forces, by age, sex and outcome), when (by year and decade), where (by department) and what civilians were doing, because these are the questions a mine action planner uses to decide where resources go first. Where civilians are hurt more often, the chance of a civilian encountering a hazard is higher, so those areas come first; work that supports the local economy, such as clearing farmland, follows. The statistics are used together with maps and an understanding of the area's terrain and geology, not on their own.

What Each Plot Shows

Figure 1 compares civilian and security-force victims per year, to show the long-term trend and when the balance between the groups changed. Figure 2 compares the share of victims killed in each group. Figure 3 ranks what civilians were doing when they were hurt, showing the unrecorded activity as its own bar. Figure 4 shows the ten departments with the most victims, split into civilians and security forces.

Results and Interpretation

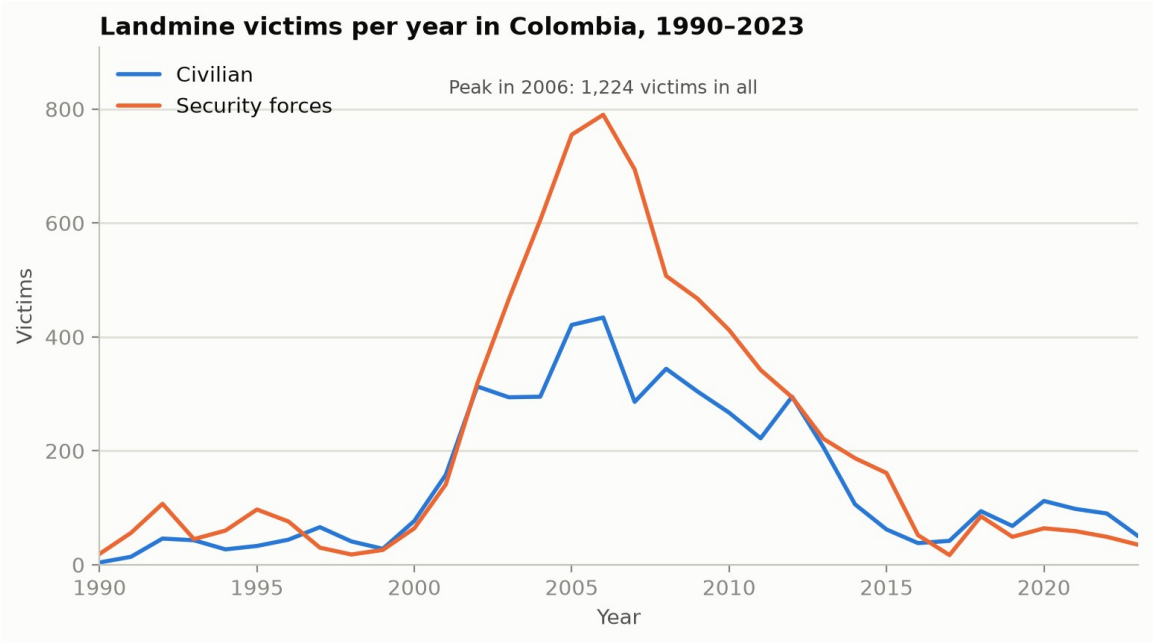
Of the 12,390 recorded victims, 59.5% were members of the security forces and 40.5% were civilians; 19.0% were killed; 10.3% were minors, all of them civilians; and 98.0% were hurt in rural areas.

Recorded victims rose sharply after 2000 and peaked in 2006 at 1,224 (790 security-force members and 434 civilians), then fell to 90 in 2016 (Figure 1). Since 2017, civilians have outnumbered security-force victims every year; in the 2020s they are 62.8% of victims, against

37.8% in the 2000s. The two yearly series are strongly correlated (r = .93), but that mainly reflects their shared rise and fall over time rather than a direct link.

Figure 1

Landmine Victims per Year in Colombia by Condition, 1990–2023



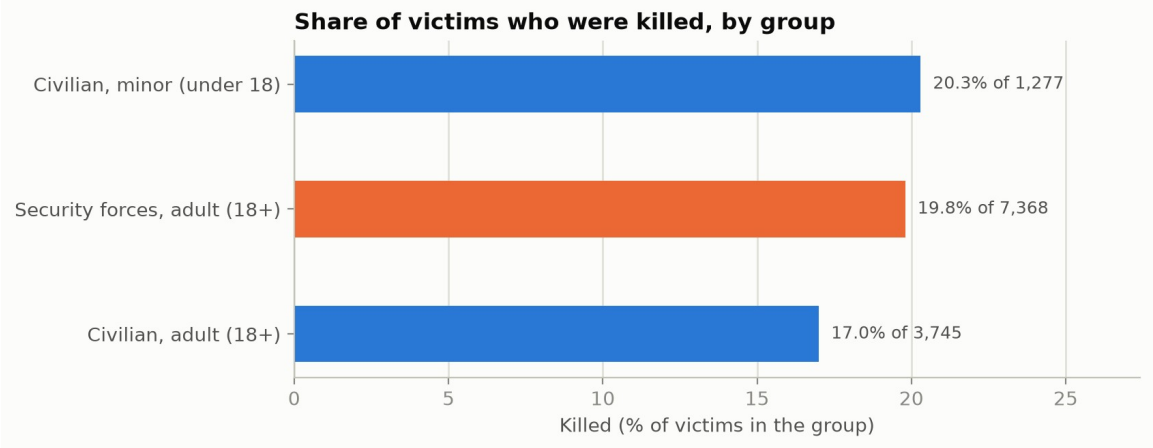
Note. Data from Presidencia de la República de Colombia (2024).

My reading of this trend comes from field experience, not from the register, which records no causes. The rise to 2006 probably reflects more fighting, or people moving into areas that were already contaminated. The fall in security-force victims since then may reflect forces being moved to other regions or tasks, away from active fighting, while drug-trafficking organisations and rebel groups continue to lay mines and booby traps, and fighting with them goes on.

About one victim in five was killed in every group: 20.3% of civilian minors, 19.8% of security-force members and 17.0% of civilian adults (Figure 2). The gaps are a few percentage points, and I did not test whether they are larger than chance would produce.

Figure 2

Share of Victims Killed, by Group



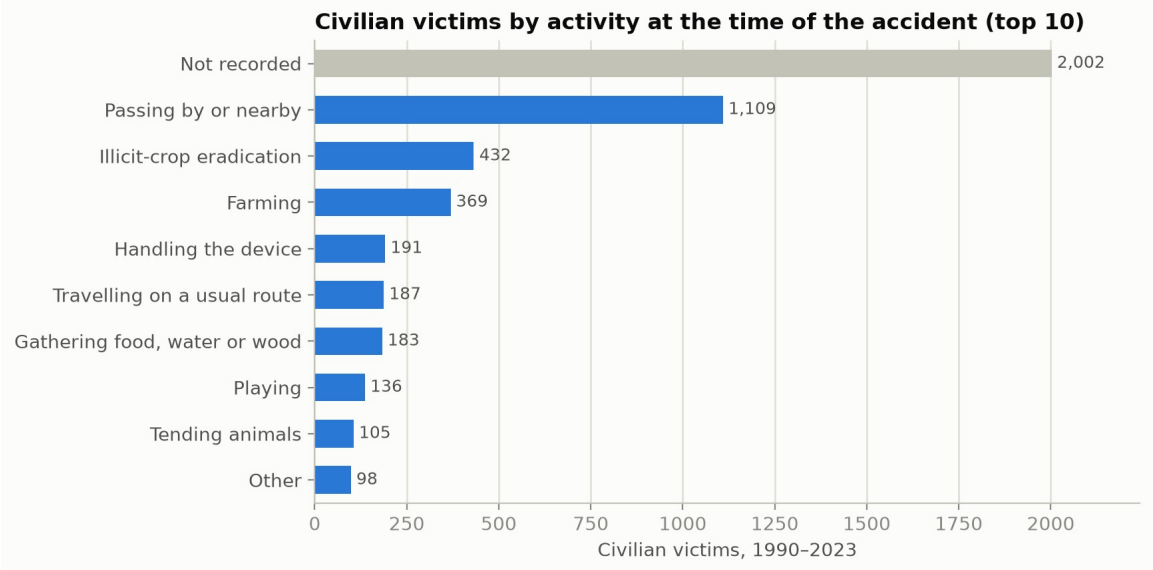
Note. No security-force victim is recorded as a minor. Data from Presidencia de la República de Colombia (2024).

In my experience as an explosive ordnance disposal technician, these mines are designed to maim rather than kill, and deaths usually follow from complications of the injury or from not reaching medical care quickly enough. First aid that stops major bleeding gives a casualty time to reach surgery. That would make the share killed depend more on access to care than on who the victim was, which fits how similar the groups are.

The activity is not recorded for 40% of civilian victims (2,002 of 5,022), so Figure 3 describes an incomplete picture. Among recorded activities, passing by or being nearby is the most common (1,109 victims), followed by illicit-crop eradication (432) and farming (369).

Figure 3

Civilian Victims by Activity at the Time of the Accident, 1990–2023 (Top 10)



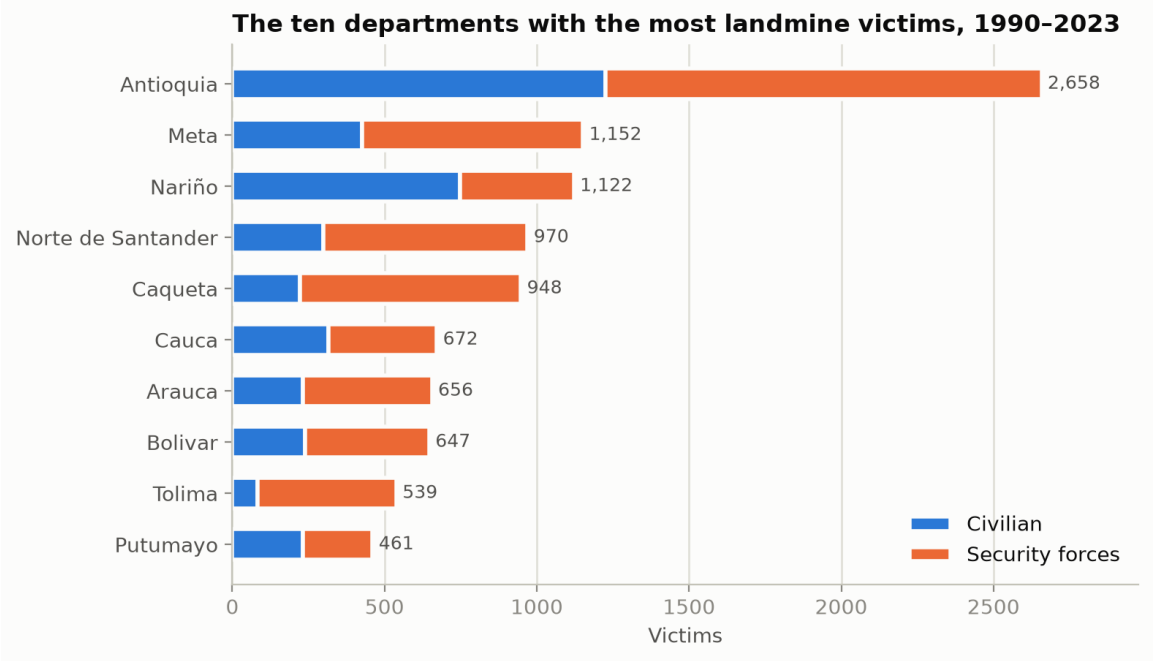
Note. "Not recorded" counts civilian victims whose activity is missing from the register. Data from Presidencia de la República de Colombia (2024).

This fits how mine action is usually prioritised in my experience: risk education first, then clearance that supports the local economy, starting with the land and infrastructure that produce income and the routes people use every day.

Antioquia has 2,658 victims, 21.5% of the national total and more than twice the next department, Meta (1,152) (Figure 4). The mix differs by department: two thirds of victims in Nariño (66.7%) were civilians, against 15.4% in Tolima and 23.4% in Caquetá.

Figure 4

The Ten Departments With the Most Landmine Victims, by Condition, 1990–2023



Note. Data from Presidencia de la República de Colombia (2024).

From my experience, several factors could explain these differences: where mines were laid relative to the supply routes of the armed forces, drug-trafficking groups and rebel groups; which groups had the means to obtain and lay mines; how each group used them, for example to protect illicit crops close to civilian communities, which raises the chance of an encounter; and the choices of the local commander. Raw totals like these should be read with maps and local knowledge before resources are allocated.

These are counts of recorded victims, and the analysis is descriptive, with no statistical tests or causal claims. The totals may reflect how completely accidents were reported, as well as where they happened.

Responsible Practice (Bias and Data Quality)

The main harm from poor cleaning would be to how big the problem looks. Mislabelled or under-counted victims can make a region look safer than it is and shift funding and clearance resources away from where they are needed. Land release itself is a detailed, deliberate process that doesn't depend on these exact numbers, but the way resources are shared between regions does.

- **Repeated rows.** The largest risk came from a routine cleaning step. A blanket de-duplication would have removed 37.4% of security-force victims but only 22.0% of civilian victims (section 3.4 of the notebook), shifting every comparison between the two groups. Removing records from some groups more than others amounts to non-random sampling of subgroups, a form of sampling bias, and models trained on biased data learn and repeat it (Mehrabi et al., 2021). I kept the rows and showed the effect in the notebook.
- **Mislabelled columns.** Read at face value, `estado` (status) and `genero` (gender) would have reported sex as a status and outcomes as a gender. The cleaning function checks the values before renaming.
- **Missing values.** Counting "Sin información" as a category would hide how much is unknown. The type of place is missing for 91.7% of victims and the activity for 40% of civilian victims, so I report those breakdowns as partial.
- **Place names.** Grouping by municipality name would merge different places; I used the official codes.
- **Privacy.** I removed free-text site names and coordinates, which this analysis does not need.

- **What I could not fix.** Data are hard to capture in war-torn areas, so under-reporting is likely, and it cannot be measured from this file. Before drawing conclusions about where risk is highest, I would compare these counts with the portal's separate table of mine events, with other sources and with maps of the area.

Reproducibility

Anyone can rerun the work from the public repository. The README gives the steps: clone the repository, create a virtual environment with Python 3.13, install the packages with `pip install -r requirements.txt`, and run every cell of `data_workflow.ipynb`. I generated `requirements.txt` with `pip freeze` in a fresh environment, so it lists the exact version of every package; a file like this lets others rebuild the same environment and avoids errors caused by later package updates (Danchev, 2022). The raw data file is in the repository, and the notebook checks its checksum on every run. The cleaned CSV is written with the same line endings on every operating system, so a rerun produces an identical file.

I used Git and GitHub to record the work, committing as each task was completed: the dataset, the requirements, ingestion, cleaning, the analysis, the charts, the summary and the README. The cleaning pipeline was developed on a separate `cleaning-pipeline` branch and merged into `main`, and both branches are on GitHub. Managing changes with a version control system is one of the practices Wilson et al. (2017) recommend. A fresh clone installed from `requirements.txt` reproduced the same outputs on Linux and on Windows, and the charts were identical pixel for pixel.

I used Claude (Anthropic). I ran the notebook and checked its results.

Sources and Citations

References

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